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Automatic glioma segmentation based on efficient U-net model using MRI images

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ABSTRACT

Gliomas are among the most aggressive and challenging brain tumors to diagnose and treat. Accurate segmentation of glioma regions in Magnetic Resonance Imaging (MRI) is essential for early diagnosis and effective treatment planning. This study proposes an optimized U-Net model tailored for glioma segmentation, addressing key challenges such as boundary delineation, computational efficiency, and generalizability. The proposed model integrates streamlined encoder-decoder pathways and optimized skip connections, achieving precise segmentation while reducing computational complexity. The model was validated on two datasets: TCGA-TCIA, containing 110 patients, and the multi-modal BraTS 2021 dataset. Comparative evaluations were conducted against state-of-the-art methods, including Attention U-Net, Trans-U-Net, DeepLabV3+, and 3D U-Net, using metrics such as Dice Coefficient, Intersection over Union (IoU), Hausdorff Distance (HD), and Structural Similarity Index (SSIM). The proposed U-Net achieved the highest performance across all metrics, with a Dice score of 92.54 %, IoU of 90.42 %, HD of 4.12 mm, and SSIM of 0.962 on the TCGA-TCIA dataset. On the BraTS dataset, it achieved comparable results, with a Dice score of 91.32 % and an IoU of 89.56 %. In contrast, other methods, such as Attention U-Net and DeepLabV3+, showed lower Dice scores of 85.62 % and 84.10 %, respectively, and higher HD values, indicating inferior boundary delineation. Additionally, the proposed model demonstrated computational efficiency, processing images in 1.5 s on average, compared to 5.0 s for Attention U-Net and 9.0 s for Trans-U-Net. These results underscore the potential of the optimized U-Net as a robust, accurate, and efficient tool for glioma segmentation. Future work will focus on clinical validation and extending the model to include automated glioma grading, further enhancing its applicability in medical imaging workflows.

1. Introduction

Gliomas, a category of brain and central nervous system tumors, are among the most aggressive and life-threatening cancers. These tumors account for a significant proportion of primary brain cancers, with survival rates heavily influenced by the stage at diagnosis and the patient's age. In the United States, individuals diagnosed with brain or central nervous system cancers face a five-year survival rate of only 36 %, underscoring the devastating nature of these malignancies. However, in younger populations, particularly children under the age of 15, the survival rate improves to approximately 75 %, reflecting the critical role of early detection and treatment [1].

The accurate identification and delineation of gliomas are paramount for prognosis, treatment planning, and therapeutic evaluation. Imaging modalities such as computed tomography (CT) and magnetic resonance imaging (MRI) have long been essential in diagnosing brain tumors. MRI, in particular, is favored for its superior contrast resolution and ability to non-invasively provide detailed information about tumor size, location, and structural abnormalities [2,3]. However, manual segmentation of glioma regions by radiologists is inherently time-consuming, prone to inter-observer variability, and inadequate for processing large datasets in clinical or research settings [4].

In recent years, advancements in artificial intelligence (AI), particularly deep learning, have transformed medical image analysis.

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Convolutional Neural Networks (CNNs) have proven highly effective in tasks ranging from image classification to segmentation, offering unparalleled accuracy in identifying complex patterns within imaging data [5]. Among CNN-based methods, Fully Convolutional Neural Networks (FCNNs) have demonstrated significant promise in biomedical image segmentation. One of the most widely adopted architectures in this domain is U-Net, which features a symmetric encoder-decoder structure complemented by skip connections that preserve spatial resolution [6–8].

Building upon the U-Net model, researchers have explored various enhancements to improve segmentation performance. For instance, Kermi et al. [9] introduced refinements to the contracting and expanding paths of the U-Net architecture, demonstrating improved accuracy in brain tumor segmentation tasks. However, challenges remain, particularly regarding the reliance on large datasets for training and the computational inefficiencies of standard U-Net implementations, which limit their feasibility for real-time clinical use.

Subsequent variants, such as Attention U-Net, incorporated attention gates to focus on relevant features while suppressing irrelevant background noise [10,11]. Trans-U-Net combined Transformer layers with U-Net to capture long-range dependencies, enhancing performance on complex segmentation tasks [12].

Other notable methods include DeepLabV3+ [13], which leverages atrous convolutions for multi-scale feature extraction, and 3D U-Net [14], designed for volumetric segmentation tasks using 3D convolutions. Despite these advancements, challenges persist, such as handling class imbalance, ensuring robustness to variations in image quality, and achieving real-time performance for clinical applications. However, these methods often involve increased computational complexity.

The objective of this study is to develop an optimized U-Net model for glioma segmentation in MRI images, addressing specific challenges such as boundary delineation, computational efficiency, and generalizability to multi-modal datasets. Unlike existing approaches that rely on complex architectures, the proposed model integrates streamlined encoder-decoder pathways, optimized skip connections, and parameter-efficient designs to achieve high segmentation accuracy with reduced computational cost. Furthermore, this study validates the model's performance on BraTS dataset, demonstrating its robustness across different glioma grades and imaging modalities.

2. Materials and methods

2.1. Data description

In this paper, the used dataset consists of a subset derived from The Cancer Genome Atlas (TCGA) and The Cancer Imaging Archive (TCIA) [15]. The data presents all modalities, resorting to FLAIR only when other modalities were unavailable. It includes data from 110 patients diagnosed with lower-grade glioma tumor (101 patients with all sequences available, and 9 patients with missing post-contrast sequence). The images were acquired with a resolution of 256 x 256 pixels, covering axial slices of the brain with a slice thickness of 1 mm. In glioma cases, contrast enhancement is particularly significant. Therefore, assessing the tumor's shape primarily relies on the abnormalities observed in Fluid-Attenuated Inversion Recovery (FLAIR) imaging. The FLAIR technique suppresses fluid signals, allowing for clearer visualization of lesions near fluid-filled spaces (Fig. 1). The dataset was divided into 70 % for training (77 patients) and 30 % for testing (33 patients). Care was taken to ensure an even distribution of tumor grades and sizes across the subsets to prevent potential biases.

2.2. Data preprocessing

All images were preprocessed to ensure consistency, including resizing to 256 x 256 pixels, intensity normalization to a range of [0, 1], and data augmentation techniques such as rotation, flipping, translation, and zooming to address class imbalance and improve model generalization.

Given the variability in tumor sizes, oversampling was performed for underrepresented tumor categories to ensure a balanced dataset. Tumor masks with smaller regions were augmented at higher rates to compensate for their scarcity. These preprocessing steps were applied uniformly to the training set, while the test set remained untouched to ensure unbiased evaluation.

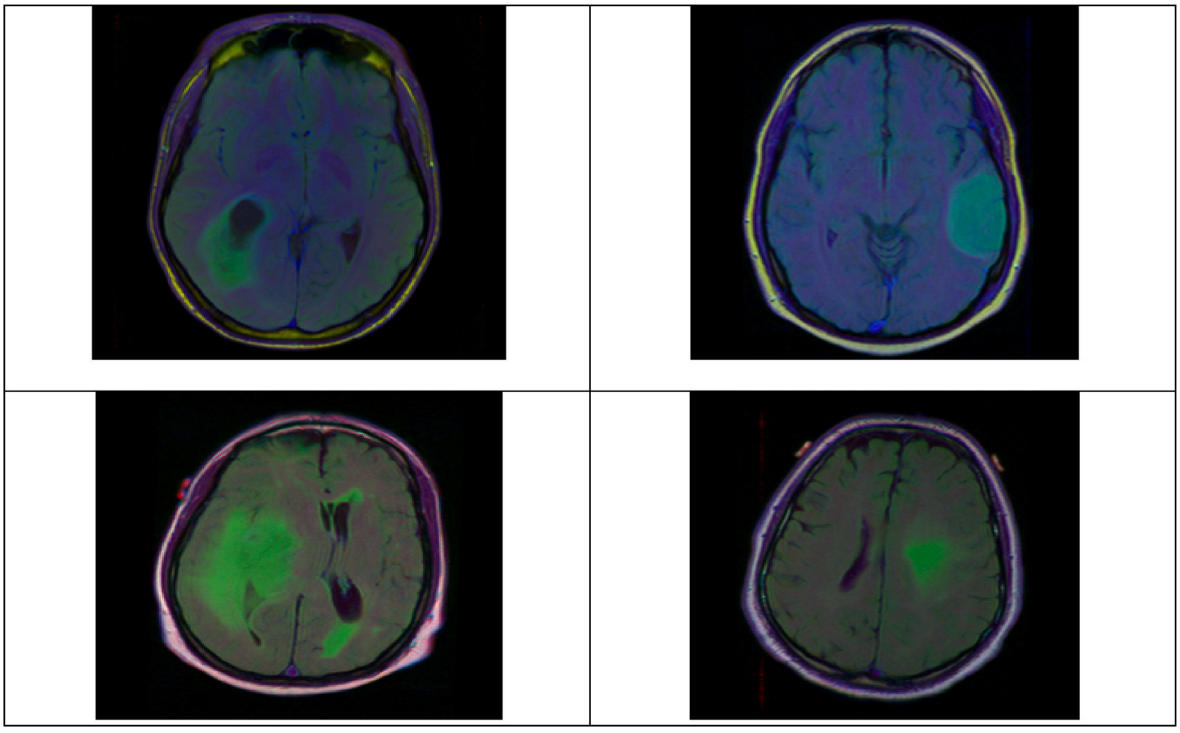


Fig 1. Examples of the employed MRI dataset.

2.3. Computational environment and resources

The experiments were performed using Python (version 3.12) as the primary programming language, with TensorFlow (version 2.16) and Keras (version 2.16) frameworks for developing and training deep learning models. Libraries such as NumPy, Pandas, Matplotlib, and Seaborn were employed for data manipulation, image processing, and evaluation tasks. The experiments were run on a laptop equipped with an Intel Core i7-1355U processor (up to 5.0 GHz, 12 MB cache), 16 GB RAM, and an NVIDIA GeForce MX550 GPU with 2 GB of memory, operating on Windows 10.

2.4. The developed U-net architecture

The Convolutional Neural Network (CNN) architecture is highly effective for the involved task of identifying and segmenting regions affected by brain cancer [16,17]. In this study, we have employed the U-Net model, a variant of CNN, for the segmentation of glioma in MRI images. The U-Net structure is particularly suitable for biomedical image segmentation due to its encoder-decoder layout and the efficient use of skip connections, which maintain critical spatial details throughout the processing pipeline.

2.4.1. Mathematical framework

In the U-Net model, segmentation is framed as a mapping function, $\mathbf{f_\theta: I \rightarrow S}$, where:

- **I** represents the set of input images within the domain $\Omega \subset R^d$ (with d $\geq$ 2 dimensions),
- **S** is the segmented output, derived as $\mathbf{S \in I^*}$, which contains the desired tumor regions in the observed image $\mathbf{I_0 \in I}$.
- $\mathbf{f_\theta}$ is a complex nonlinear function parameterized by $\boldsymbol{\theta}$, optimized during training to minimize a loss function and yield high-accuracy segmentation.

In the training phase, the model's parameters, encapsulated in $\boldsymbol{\theta}$, are adjusted to minimize the error between the predicted segmentation and the ground truth. The function $\mathbf{f_\theta}$ represents a series of transformations achieved through convolutional layers, arranged to maximize feature extraction and minimize computational expense. This multi-layer structure allows high-level feature extraction without requiring fully connected layers, thereby enhancing segmentation performance while maintaining efficiency [18].

For further details on segmentation model principles, refer to previous research on segmentation architectures [19]. The CNN method has been widely employed for brain cancer analysis with a focus on deep networks [20]. Pereira et al. [21] successfully identified brain cancer regions using CNN on MRI datasets, leveraging convolutional kernels and stacked convolutional layers. In this study, our U-Net approach, optimized for glioma detection, effectively captures complex data patterns from MRI images [22,23].

2.4.2. U-net architecture design

The U-Net model architecture used for glioma segmentation follows these main components:

1. **Input MRI Image**: The model takes an MRI image as input.
2. Encoder:
   - The encoder is responsible for detecting the context of the image by down sampling it through a series of convolutional and max-pooling layers.
   - **Convolutional Layers**: Each convolutional layer applies a set of filters to the input image, which helps in capturing spatial features such as boundary and textures.
   - **Max-Pooling Layers**: After each convolutional block, a max-pooling operation reduces the spatial dimensions of the feature maps, retaining the most prominent features and allowing the model to detect more information about the input image at different scales. This process is repeated over several levels, effectively transforming the input image into a dense representation that encodes both local and global patterns relevant to glioma regions.
3. Decoder:
   - The decoder reconstructs the spatial dimensions by progressively upsampling the feature maps back to the original input size.
   - **Deconvolution (or Transposed Convolution) Layers**: These layers upsample the feature maps, reversing the effect of down-sampling done by the encoder.
   - **Up-Pooling Layers**: Up-pooling layers restore the spatial resolution of the feature maps to match the input image size.
   - The upsampled feature maps are combined with corresponding feature maps from the encoder part via skip connections, which helps retain fine details from the original input image.
   - **Skip Connections**: To preserve spatial context, feature maps from the encoding path are concatenated with corresponding maps in the decoding path, facilitating accurate localization.
4. **Output Segmented MRI Image**: The final layer produces a binary image that highlights the segmented glioma ROI (Region of Interest). This output mask highlights gliomas regions within the MRI images, as depicted in Fig. 2.

2.4.3. Training and hyperparameters

The Developed U-Net model was trained on The Cancer Genome Atlas (TCGA) dataset, with specific training parameters shown in Table 1. During training, several configurations were evaluated to optimize segmentation performance, such as adjusting the initial learning rate ε, batch size, and the dropout probability is set to 0.2. For the bias learned during training, it is part of the neuron's pre-activation parameters.

The Leaky ReLU (Rectified Linear Unit) function is a variant of the standard ReLU activation function, and its equation is defined as:

$$f(x) = \max(\alpha x, x) \tag{1}$$

For each convolutional layer, the Leaky ReLU activation function was applied with a constant $\alpha = 0.01$ to control the negative slope of the input x, aiding in convergence by maintaining the model's ability to learn complex features.

2.5. Evaluation metrics

The performance of the proposed U-Net model was evaluated using the following metrics:

- Dice Coefficient: Measures the overlap between predicted and ground truth segmentations, with higher values indicating better performance.
- Intersection over Union (IoU): Assesses the ratio of the intersection to the union of predicted and ground truth segmentations.
- Hausdorff Distance (HD): Quantifies the maximum distance between the predicted and ground truth boundaries, providing insight into boundary precision.
- Structural Similarity Index (SSIM): Evaluates perceptual similarity between predicted and ground truth images by considering luminance, contrast, and structural information.

2.6. Additional validation on BraTS dataset

To ensure the generalizability of the proposed U-Net model, we validated its performance on the BraTS 2021 dataset [24]. This dataset contains multi-modal MRI scans (T1, T1-contrast enhanced, T2, and FLAIR) of lower-grade gliomas. For consistency with our primary

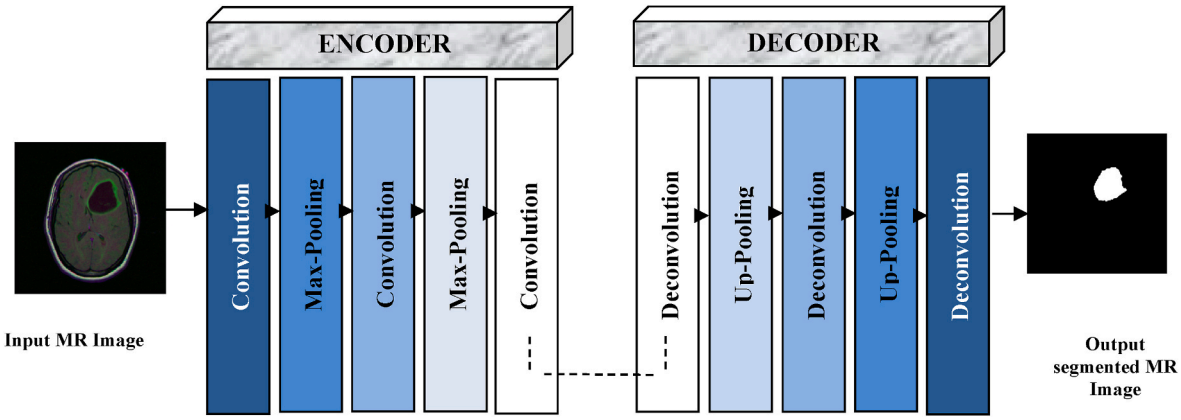


Fig 2. U-net Encoder-Decoder structure

Table 1
The hyper parameters of the used method.

Initialization	Leaky ReLU	Dropout	Training			
Bias	**α**	**p**	**Initial ε**	**Final ε**	**Epochs**	**Batch**
0.1	0.01	0.2	0.003	0.00003	100	128

dataset, only the FLAIR modality was used in this validation. The images were resized to 256 x 256 pixels and preprocessed similarly to the TCGA dataset, including intensity normalization and data augmentation. The validation subset consisted of 125 cases, and segmentation performance was evaluated using Dice, IoU, Structural Similarity Index (SSIM), and Hausdorff Distance (HD) metrics.

2.7. Comparative approach of automatic glioma ROI segmentation

The proposed model was evaluated against Attention U-Net, Trans-U-Net, DeepLabV3+, and 3D U-Net, well-known methods in medical image segmentation. These models were implemented using TensorFlow and trained under identical conditions to ensure a fair comparison. Performance metrics, including Dice Coefficient, Intersection over Union (IoU), and Hausdorff Distance, and Structural Similarity Index (SSIM) were calculated for all models.

To validate the statistical significance of the performance differences, paired t-tests were conducted for Dice and IoU metrics, with a significance level of $p < 0.05$.

2.7.1. Attention U-Net

The Attention U-Net model [10], presented here as a comparison method, is designed to enhance segmentation accuracy by incorporating attention mechanisms. This model has shown promising results in medical imaging and is used here to benchmark the performance of our proposed U-Net-based approach.

In abdominal CT images, the attention U-Net [11] was designed for multi-class segmentation. Moreover, the system employed was adapted for pneumonia segmentation. Four down and up-sampling levels are employed after the concatenation step. The ReLU activation function is used in batch normalization. Additionally, additive attention and ReLU attention activation are used. Indeed, stridden convolution layers and bilinear interpolation were used for down-sampling and up-sampling, respectively (see Fig. 3).

2.7.2. Trans-U-Net

Similarly, the Trans-U-Net model, another comparison method, integrates Transformer layers with U-Net to leverage both spatial and contextual information. This architecture is included for comparison to highlight the relative advantages of our proposed model in segmenting glioma regions.

In therapeutic applications, the Trans-U-Net [12] model has been used for synapse multi-organs segmentation into 8 abdominal organ categories based on abdominal CT images; as shown in Fig. 4. Compared to different-organs segmentation, the pneumonia segmentation task involves predicting a binary mask instead of multiclass pixel-wise segmentation. The Trans-U-Net architecture is adapted by combining the U-Net and Transformer systems.

2.7.3. 3D U-net

The 3D U-Net architecture extends the 2D U-Net design into three dimensions, making it ideal for processing volumetric medical imaging data, such as MRI and CT scans. By using 3D convolutions and 3D pooling layers, the model effectively captures volumetric context, which is critical for segmenting three-dimensional structures like brain tumors. Skip connections between encoder and decoder layers ensure the preservation of spatial details, enabling precise boundary delineation. Despite its high computational cost, 3D U-Net remains a powerful tool for volumetric segmentation tasks, particularly in applications where the spatial relationship between slices is critical [14].

2.7.4. DeepLabV3+

DeepLabV3+ builds on the semantic segmentation capabilities of DeepLabV3 by introducing an encoder-decoder architecture. This model combines atrous spatial pyramid pooling (ASPP) for multi-scale feature extraction with a decoder to refine the segmentation output. The ASPP module utilizes dilated convolutions with varying dilation rates, enabling the model to capture features at multiple scales. The addition of a decoder improves the preservation of spatial details, particularly at object boundaries, making DeepLabV3+ effective for segmenting complex medical structures such as gliomas. Its modular design allows for flexibility in handling high-resolution images with minimal computational overhead, positioning it as a benchmark for segmentation tasks in medical imaging [13].

3. Results and discussions

In the present study, The proposed U-Net model achieved the highest Dice Coefficient (92.54 %) and IoU (90.42 %) compared to Attention U-Net (85.62 % Dice, 82.53 % IoU), Trans-U-Net (81.24 % Dice, 77.39 % IoU), DeepLabV3+ (84.10 % Dice, 81.12 % IoU), and 3D U-Net (86.50 % Dice, 84.30 % IoU) (Table 2). Hausdorff Distance analysis further highlighted the model's superior boundary delineation capabilities, with the proposed U-Net achieving the lowest value of 4.12 mm. The proposed method showed also a SSIM score of 0.962 which demonstrates its capability to produce segmentations that closely resemble ground truth images in terms of structural and perceptual similarity.

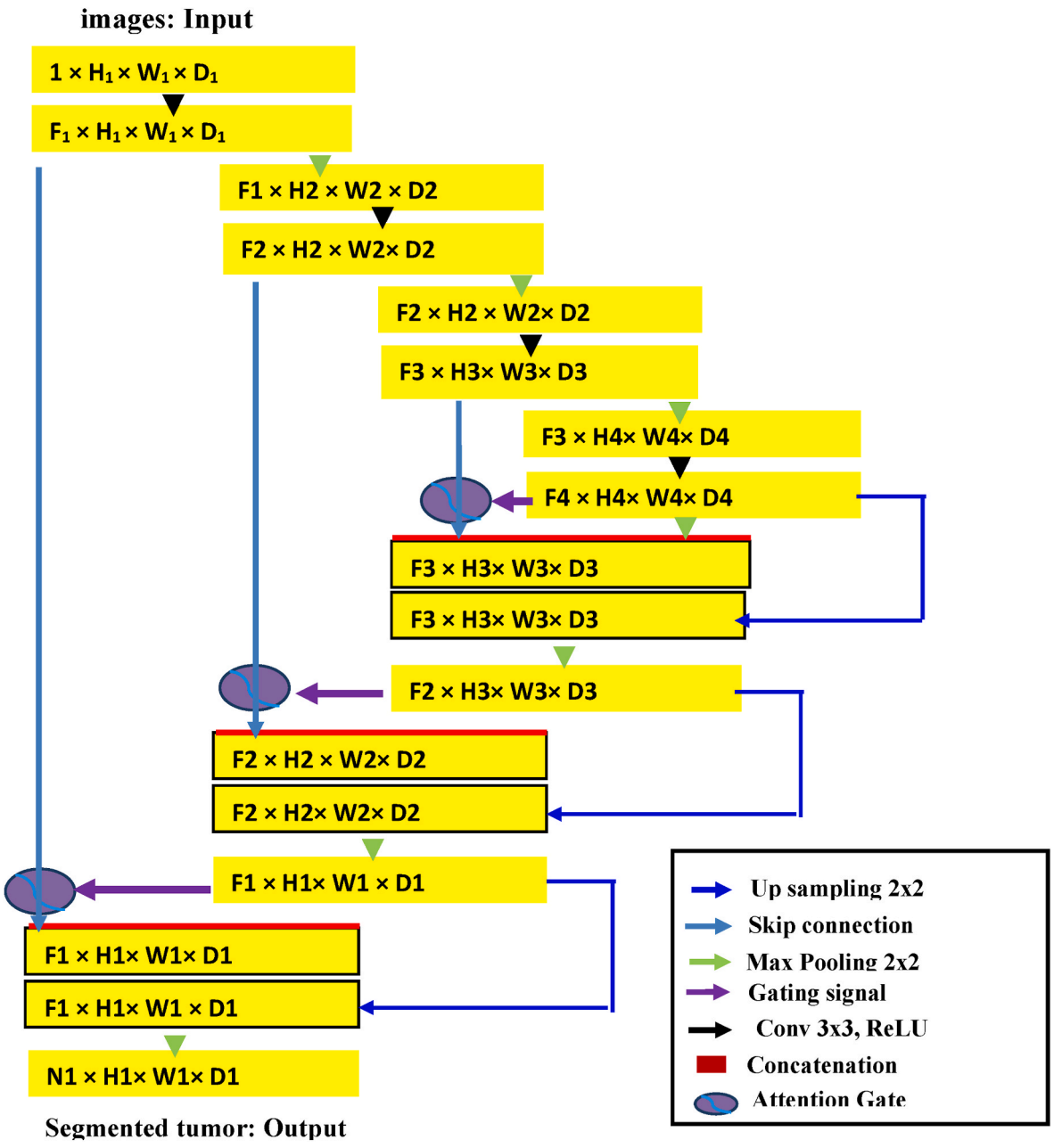


Fig 3. Attention U-Net architecture.

Paired t-tests confirmed that the performance differences between the proposed U-Net and other models were statistically significant ($p < 0.05$). These results substantiate the robustness and accuracy of the proposed model across diverse glioma segmentation tasks. The superior performance is attributed to the streamlined encoder-decoder design, optimized skip connections, and reduced parameter complexity, which allow for efficient feature extraction and spatial detail preservation.

The IoU score further confirms this advantage by demonstrating the model's ability to accurately segment glioma areas with minimal false positives and false negatives. Compared to the comparative methods, the modified U-Net architecture optimized parameters allowing efficient feature extraction without the additional complexity of attention or transformer layers. Overall, these results underscore the effectiveness of the developed U-Net model in delivering accurate and reliable segmentation outcomes for glioma detection in MRI images.

Fig. 5 illustrates the original and segmented MRI images to present the effectiveness of the employed approach. The current segmentation results are attained with respect to the accuracy of the segmentation stage using 30 % from the MRI dataset.

The computation time of the Developed U-Net model was thoroughly evaluated and compared with traditional Trans-U-Net and Attention-U-Net models. Results showed that the proposed U-Net model achieved significantly better efficiency, processing each image in 1.5 s on average. This performance was notably faster than the comparative methods; Trans-U-Net, Attention-U-Net, DeepLabV3+, and 3D U-Net, which required 5, 9, 6.5, and 8 s per image, respectively, due to their more complex architectures. The computational efficiency of the proposed U-Net model was achieved through a combination of reduced parameter count, streamlined encoder-decoder pathways, and optimized skip connections. These enhancements reduced computational overhead while maintaining high segmentation accuracy, making the model more suitable for real-time applications. These findings indicate that our model achieves a balance between segmentation accuracy and processing efficiency, addressing a critical need in glioma ROI analysis.

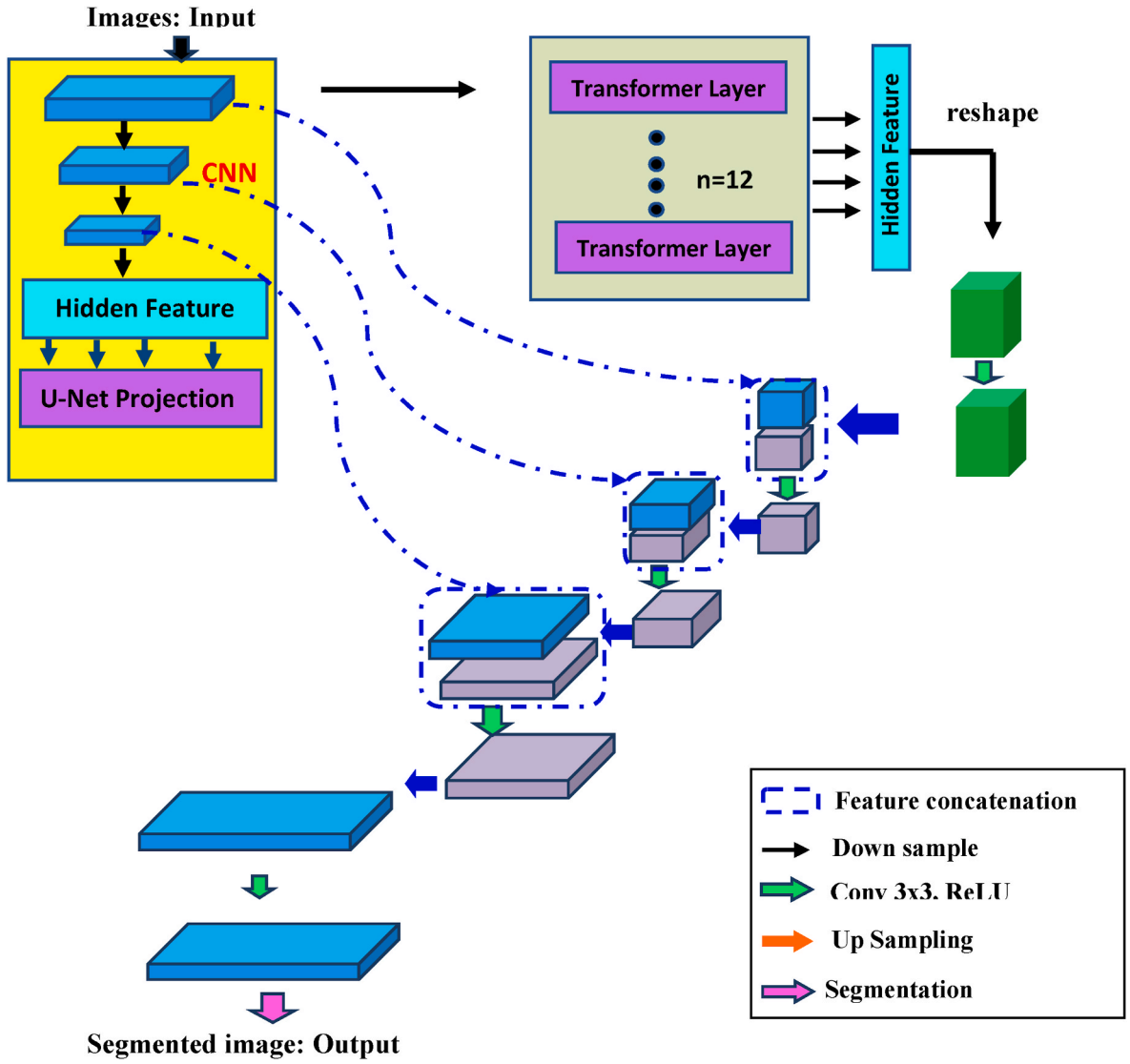


Fig 4. Trans-U-Net architecture.

Table 2
Performance metrics across models.

Model	Dice Coefficient (%)	IoU (%)	Hausdorff Distance (mm)	Processing Time (s)	Structural Similarity Index: SSIM
Proposed U-Net	92.54	90.42	4.12	1.5	0.962
Attention U-Net	85.62	82.53	6.10	5.0	0.928
Trans-U-Net	81.24	77.39	6.80	9.0	0.910
DeepLabV3+	84.10	81.12	5.50	6.5	0.922
3D U-Net	86.50	84.30	5.00	8.0	0.935

The validation of the proposed U-Net model on the BraTS 2021 dataset yielded comparable results to those achieved on the TCGA dataset. The model achieved a Dice score of 91.87 % and an IoU of 89.94 %, demonstrating high segmentation accuracy. The Hausdorff Distance of 4.12 mm indicates precise boundary delineation, and the average processing time of 1.6 s per image highlights the model's computational efficiency. These results are consistent with those obtained on the primary dataset, where the Dice score was 92.54 % and the IoU score was 90.42 %. These results validate the generalizability of the proposed U-Net across different datasets and show its ability to perform well with multi-modal MRI data.

The preprocessing pipeline played a critical role in achieving the reported performance of the proposed U-Net model. Normalization ensured consistent input data quality, allowing the model to focus on feature extraction rather than intensity variations. Data augmentation improved the model's ability to generalize by simulating real-world variations, such as different tumor orientations, sizes, and anatomical distortions. Addressing class imbalances by oversampling and augmenting smaller tumor regions contributed to the model's accuracy in segmenting both small and large tumors effectively. These preprocessing steps not only enhanced the model's robustness but also improved its reproducibility, enabling its application to datasets with similar characteristics.

The U-Net architecture [25] has inspired numerous variants tailored for specific segmentation tasks, as demonstrated by recent advancements. Azad et al. [26] provided a comprehensive review of U-Net's success in medical imaging, emphasizing the impact of attention mechanisms and multi-scale designs. Wu et al. [27] proposed the UIU-Net, a nested U-Net model that excels in detecting small objects, highlighting the importance of hierarchical feature extraction. Zhang et al. [28] introduced BCU-Net, which integrates ConvNeXt and U-Net to combine the benefits of convolutional and transformer-based

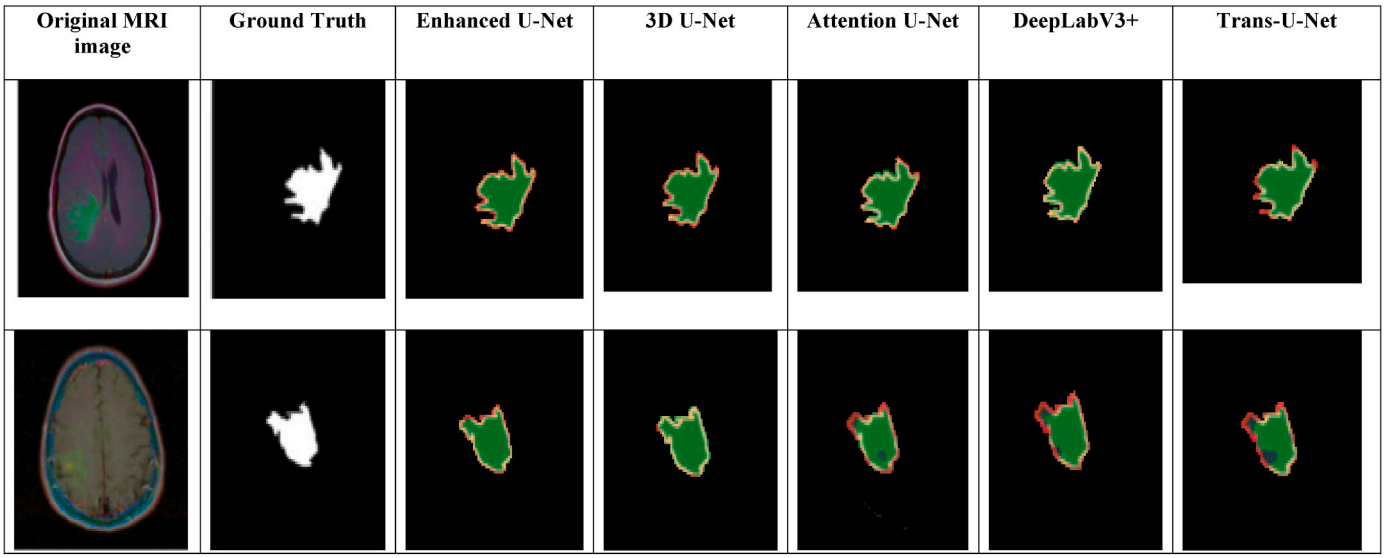


Fig 5. Visualization results of the Enhanced UNet and other network models in the glioma Dataset segmentation task.

architectures, enhancing segmentation accuracy. Zhao et al. [29] presented NAU-Net, incorporating neighbor attention mechanisms to improve feature selection, achieving remarkable results in diabetic retinopathy segmentation. These studies underscore the versatility of U-Net and its derivatives, providing valuable insights that guided the optimization of our proposed model.

Compared to recent U-Net advancements such as UIU-Net, BCU-Net, and NAU-Net, our proposed model focuses on optimizing computational efficiency while maintaining high segmentation accuracy. By simplifying the encoder-decoder pathways and incorporating refined skip connections, our approach ensures robust performance without the complexity of attention or nested mechanisms, making it suitable for clinical applications.

The proposed U-Net model incorporates several architectural enhancements tailored to address the unique challenges of glioma segmentation. One critical challenge is the accurate delineation of tumor boundaries, which often exhibit irregular shapes and complex textures. To address this, the model utilizes optimized skip connections that enable the decoder to incorporate fine-grained spatial details from earlier encoder layers. These connections ensure that critical boundary information is preserved during the upsampling process, leading to more precise segmentation maps. Additionally, the encoder-decoder pathways were streamlined by reducing redundant layers, which improved computational efficiency while retaining the model's ability to capture hierarchical features. This optimization also mitigates the risk of overfitting, especially when working with relatively small datasets like ours. The use of Leaky ReLU activation ($\alpha = 0.01$) further enhances gradient flow, reducing the likelihood of vanishing gradients during training. Collectively, these modifications enhance the model's capacity to handle the spatial variability of gliomas, improve segmentation accuracy, and achieve faster processing times compared to other methods.

The performance of the proposed U-Net model aligns with findings from recent studies that emphasize the importance of hybrid methods and optimization techniques in brain tumor segmentation. Saifullah and Drężewski demonstrated significant improvements using Particle Swarm Optimization-enhanced preprocessing (PSO) and ensemble learning methods [30–32]. These approaches, combined with deep learning frameworks such as U-Net and DeepLabV3+, highlight the value of optimization strategies for segmentation tasks. Our model's improvements, achieved through streamlined encoder-decoder pathways and optimized skip connections, complement these strategies by providing a lightweight and efficient alternative while maintaining high segmentation accuracy.

4. Conclusion

The identification of the glioma region poses a crucial challenge in the assessment of tumor grade. The CNN methodology is noted for achieving the detection of glioma ROI. Multiple methods deal with problems related to the vanishing gradient during training. Therefore, in order to address these segmentation issues, this study employs the U-Net framework. The proposed segmentation approach demonstrates both the shortest computation time and superior dice and IoU values when compared to current methods from the literature for gliomas segmentation using MRI images. However, there are several directions for future research to enhance the model's applicability and robustness further. First, we plan to extend the model to detect both low-grade and high-grade gliomas by training and validating it on larger, multi-center datasets. Incorporating a diverse range of tumor grades will improve the model's generalizability and ensure that it can handle various glioma subtypes effectively. Additionally, collaborations with radiologists and clinicians will be established to evaluate the model's performance in real-world scenarios. This assessment will focus on the model's utility for diagnosis, treatment planning, and monitoring, ensuring its practical relevance and acceptance in clinical practice.

CRediT authorship contribution statement

Yessine Amri: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Amine Ben Slama:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Zouhair Mbarki:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Ridha Selmi:** Writing – review & editing, Visualization, Methodology. **Hedi Trabelsi:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Ethics approval and consent to participate

This study was conducted following the principles of the Declaration of Helsinki developed by the World Medical Association, as established by the World Medical Association, and received approval from the Human Ethics committee of Béchir Hamza Children's Hospital of Tunis.

All procedures were carried out in compliance with relevant guidelines and regulations.

The requirement for informed consent was waived by Bechir Hamza Children's Hospital Human Ethics committee.

Consent for publication

Not Applicable.

Availability of data and materials

For Data evaluation please contact Dr. Ben Slama Amine at; amine.benslama@istmt.utm.tn.

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Declaration of competing interest

The authors declare that they have no known competing financialinterestsor personal relationships that could have appeared to influence the work reported in this paper.

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